Evaluating the Transport Impact of Large Post-Quantum DNSSEC Signatures

Theoretical Part (Project – Part 1)

Justin Kasper
Student no. 145971
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Mert Duman
Student no. 145872
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Efim Shchegolev
Student no. 144376
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

Sezer Tetik
Student no. 145873
ISCTE – Instituto Universitário de Lisboa
Lisbon, Portugal

***Abstract*—Post-quantum signature schemes such as ML-DSA-44, standardized by NIST in FIPS 204, produce signatures and public keys that are one to two orders of magnitude larger than those of the classical algorithms used in DNSSEC today. As a result, signed DNS responses routinely exceed the 1,232-byte EDNS(0) payload recommendation and the typical Internet MTU, which pushes them onto IP fragmentation, truncation with TCP fallback, or alternative transports. This theoretical part reviews the DNS and DNSSEC foundations, the classical and post-quantum signature algorithms relevant to DNSSEC, and the message-size and transport mechanisms that govern how large responses are delivered, including TCP fallback, Application-layer Request-based Resource Record Fragmentation (ARRF), and DNS over QUIC (DoQ). It then surveys related work and identifies the gap addressed by the practical part of the project: a controlled, reproducible comparison of these transports for ML-DSA-44 against an ECDSA baseline under varying latency and packet loss.**

***Index Terms*—DNSSEC, post-quantum cryptography, ML-DSA, TCP fallback, IP fragmentation, ARRF, DNS over QUIC**

I. Introduction

The Domain Name System (DNS) translates human-readable domain names into the numerical addresses that are required for almost every transaction on the Internet [1]. Despite this critical role, the original DNS protocol was not designed with any mechanism for verifying that a response actually originates from the authoritative source or that it has not been altered in transit [2]. The DNS Security Extensions (DNSSEC) were introduced to close this gap by adding data origin authentication and data integrity protection to DNS responses through digital signatures, without, however, providing confidentiality [2]–[4]. More than two decades after its initial specification, DNSSEC is considered the best current practice for authenticating DNS data [5], even though deployment remains uneven: recent estimates suggest that fewer than ten percent of website domain names are signed and that only around a third of queries to recursive resolvers are actually validated [5].

The cryptographic foundation of DNSSEC, however, is now under long-term threat. The signature algorithms that are most widely deployed today, in particular RSA and the Elliptic Curve Digital Signature Algorithm (ECDSA) [6], [7], derive their security from the computational hardness of integer factorization and the discrete logarithm problem. A sufficiently large, cryptographically relevant quantum computer would be able to solve both of these problems efficiently using Shor's algorithm [8], which would allow an attacker to forge DNSSEC signatures and therefore to present spoofed DNS data as authentic.[1] To address this risk, the U.S. National Institute of Standards and Technology (NIST) has standardized the Module-Lattice-Based Digital Signature Algorithm (ML-DSA) in FIPS 204, a signature scheme whose security relies on the hardness of lattice problems and that is believed to remain secure even against an adversary equipped with a quantum computer [9].

Post-quantum cryptography (PQC) schemes such as ML-DSA, however, do not come without cost. The smallest standardized ML-DSA parameter set, ML-DSA-44, produces signatures of 2,420 octets and public keys of 1,312 octets [9], [10], which is one to two orders of magnitude larger than a typical ECDSA signature. For comparison, an ECDSA P-256 signature occupies only 64 bytes, so an ML-DSA-44 signature is almost 38 times larger [11]. When such a signature is embedded into a DNSSEC response together with the DNS

Network and Information Systems Security, ISCTE – Instituto Universitário de Lisboa. Submitted on 28 October 2026.

[1]This is a long-term, not an immediate, threat: at the time of writing, no cryptographically relevant quantum computer is publicly known to exist. Nevertheless, migration planning for long-lived infrastructure such as DNS is widely recommended well in advance of such a scenario.

header, resource records, and—where applicable—a DNSKEY record, the resulting message can easily exceed the conservative 1,232-byte UDP payload size that is commonly recommended in operational practice today, and it will frequently exceed the roughly 1,500-byte Maximum Transmission Unit (MTU) of a typical Ethernet path [12], [13]. Because DNS was originally designed around a 512-byte UDP message size [1] and later extended via EDNS(0) to negotiate larger UDP payloads [12], oversized post-quantum DNSSEC responses risk falling back to one of two imperfect mechanisms: IP-layer fragmentation, which is unreliable because many firewalls and other middleboxes silently drop fragmented packets [14], [15], or truncation of the UDP response followed by a retry over TCP, which guarantees delivery but adds at least one additional round trip and extra server-side state [14]. A drop-in replacement of classical signatures with quantum-safe ones is expected to increase DNSSEC query resolution times by at least a factor of two: a response carrying large post-quantum signatures is likely to be truncated, which wastes the initial UDP round trip, and the subsequent retry over TCP incurs at least two further round trips because of the three-way handshake [16].

The DNS operator community has already begun to experiment with mitigations for this problem. Cloudflare has deployed experimental ML-DSA-44 validation on its public 1.1.1.1 resolver, relying on TCP fallback whenever a response becomes oversized [11], and an open-source extension of the CoreDNS authoritative server has demonstrated on-the-fly signing with several PQC signature families, including ML-DSA [17], [18]. At the same time, the research literature has proposed alternative transport mechanisms that try to avoid the drawbacks of both IP fragmentation and plain TCP fallback. Application-layer Request-based Resource Record Fragmentation (ARRF) fragments oversized DNS resource records at the DNS protocol layer itself rather than relying on the IP layer, with the resolver requesting any missing fragments explicitly [19]. DNS over QUIC (DoQ) instead moves the entire DNS exchange onto a dedicated, encrypted QUIC connection, which has its own packetization and loss-recovery behavior and therefore sidesteps classical IP fragmentation entirely [20].

Although each of these mechanisms—plain TCP fallback, ARRF, and DoQ—has been described or prototyped individually, a direct, controlled, and reproducible comparison of their latency, bandwidth, and reliability behavior specifically under the conditions created by post-quantum DNSSEC signatures is still largely missing from the public literature. This gap motivates the present project. Using ECDSA-based DNSSEC as a classical baseline and an ML-DSA-44-based DNSSEC deployment as the post-quantum case, this project evaluates DNS resolution under simulated network conditions with varying latency and packet loss, measuring query latency, resource utilization, bandwidth consumption, packet sizes, and the frequency of TCP fallback. Time permitting, the evaluation is extended to ARRF-based fragmentation and to DNS over QUIC, in order to assess which combination of signature algorithm and transport mechanism currently represents the most realistic path toward a quantum-resistant DNSSEC deployment.

Concretely, the project is guided by the following research questions, which will be answered in the practical part (Project – Part 2):

RQ1: How much do ML-DSA-44 signatures increase DNSSEC response sizes, packet sizes, and bandwidth consumption, and how often do they trigger truncation and TCP fallback, compared with the ECDSA baseline?

RQ2: How do the end-to-end query latency and resource utilization of post-quantum DNSSEC with TCP fallback change as path latency and packet loss increase?

RQ3: (Time permitting.) To what extent can ARRF or DNS over QUIC reduce the overhead identified in RQ1 and RQ2?

The remainder of this (theoretical) part of the report is structured as follows. Section **??** introduces the theoretical foundations that are necessary to understand the problem: the DNS and DNSSEC protocols, the classical and post-quantum signature algorithms relevant to this project, and the mechanisms that govern DNS message size and transport selection. Section **??** positions this project with respect to the existing state of the art and related work. Section **??** closes the theoretical part with a brief summary and a transition toward the practical implementation and evaluation that are reported in Part 2 of this project.

References

[1] P. Mockapetris, "Domain names—Implementation and specification," RFC 1035, STD 13, Internet Engineering Task Force (IETF), Nov. 1987. [Online]. Available: https://www.rfc-editor.org/rfc/rfc1035. [Accessed: Oct. 20, 2026].

[2] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "DNS security introduction and requirements," RFC 4033, Internet Engineering Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4033. [Accessed: Oct. 20, 2026].

[3] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "Resource records for the DNS security extensions," RFC 4034, Internet Engineering Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4034. [Accessed: Oct. 20, 2026].

[4] R. Arends, R. Austein, M. Larson, D. Massey, and S. Rose, "Protocol modifications for the DNS security extensions," RFC 4035, Internet Engineering Task Force (IETF), Mar. 2005. [Online]. Available: https://www.rfc-editor.org/rfc/rfc4035. [Accessed: Oct. 20, 2026].

[5] P. Hoffman, "DNS security extensions (DNSSEC)," RFC 9364, BCP 237, Internet Engineering Task Force (IETF), Feb. 2023. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9364. [Accessed: Oct. 20, 2026].

[6] P. Hoffman and W. C. A. Wijngaards, "Elliptic curve digital signature algorithm (DSA) for DNSSEC," RFC 6605, Internet Engineering Task Force (IETF), Apr. 2012. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6605. [Accessed: Oct. 20, 2026].

[7] P. Wouters and O. Sury, "Algorithm implementation requirements and usage guidance for DNSSEC," RFC 8624, Internet Engineering Task Force (IETF), Jun. 2019. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8624. [Accessed: Oct. 20, 2026].

[8] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM J. Comput.*, vol. 26, no. 5, pp. 1484–1509, 1997, doi: 10.1137/S0097539795293172.

[9] National Institute of Standards and Technology, *Module-Lattice-Based Digital Signature Standard*, FIPS 204, Aug. 2024, doi: 10.6028/NIST.FIPS.204. [Online]. Available: https://csrc.nist.gov/pubs/fips/204/final. [Accessed: Oct. 20, 2026].

[10] B. E. Westerbaan and S. Schmieg, "Module-lattice digital signature algorithm for DNSSEC," IETF Internet-Draft draft-westerbaan-dnssec-mldsa-04 (work in progress), Aug. 2026. [Online]. Available: https://datatracker.ietf.org/doc/draft-westerbaan-dnssec-mldsa/. [Accessed: Oct. 20, 2026].

[11] S. Neuteboom and B. Westerbaan, "1.1.1.1 now supports post-quantum DNSSEC, all 2,420 bytes of it," Cloudflare Blog, Sep. 10, 2026. [Online]. Available: https://blog.cloudflare.com/post-quantum-dnssec-1111/. [Accessed: Oct. 2, 2026].

[12] J. Damas, M. Graff, and P. Vixie, "Extension mechanisms for DNS (EDNS(0))," RFC 6891, STD 75, Internet Engineering Task Force (IETF), Apr. 2013. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6891. [Accessed: Oct. 20, 2026].

[13] Cloudflare, "Maximum transmission unit and maximum segment size," Cloudflare WAN Documentation, Aug. 24, 2026. [Online]. Available: https://developers.cloudflare.com/cloudflare-wan/reference/mtu-mss/. [Accessed: Oct. 2, 2026].

[14] J. Dickinson, S. Dickinson, R. Bellis, A. Mankin, and D. Wessels, "DNS transport over TCP—Implementation requirements," RFC 7766, Internet Engineering Task Force (IETF), Mar. 2016. [Online]. Available: https://www.rfc-editor.org/rfc/rfc7766. [Accessed: Oct. 20, 2026].

[15] A. Herzberg and H. Shulman, "Fragmentation considered poisonous, or: One-domain-to-rule-them-all.org," in *Proc. 2013 IEEE Conf. Communications and Network Security (CNS)*, 2013, pp. 224–232, doi: 10.1109/CNS.2013.6682711.

[16] A. S. Rawat and M. P. Jhanwar, "Post-quantum DNSSEC with faster TCP fallbacks," in *Progress in Cryptology – INDOCRYPT 2024*, Lecture Notes in Computer Science. Cham, Switzerland: Springer, 2024, pp. 212–236, doi: 10.1007/978-3-031-80311-6_11.

[17] J. Gento Suela, J. Blanco-Romero, F. Almenares Mendoza, and D. Díaz-Sánchez, "Implementing and evaluating post-quantum DNSSEC in CoreDNS," in *Proc. 2025 Int. Conf. Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, 2025, pp. 420–427, doi: 10.1109/MSWIM67937.2025.11308970.

[18] CoreDNS, "CoreDNS," GitHub repository. [Online]. Available: https://github.com/coredns/coredns. [Accessed: Oct. 2, 2026].

[19] J. Goertzen and D. Stebila, "Post-quantum signatures in DNSSEC via request-based fragmentation," in *Proc. 14th Int. Conf. Post-Quantum Cryptography (PQCrypto 2023)*, Lecture Notes in Computer Science, vol. 14154, 2023, pp. 535–564, doi: 10.1007/978-3-031-40003-2_20.

[20] C. Huitema, S. Dickinson, and A. Mankin, "DNS over dedicated QUIC connections," RFC 9250, Internet Engineering Task Force (IETF), May 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9250. [Accessed: Oct. 2, 2026].